DIGITAL SIGNAL PROCESSING

Date: 21/01/2025

Sun Mon **Tue** Wed Thu Fri Sat

DAY - 1

Applications:

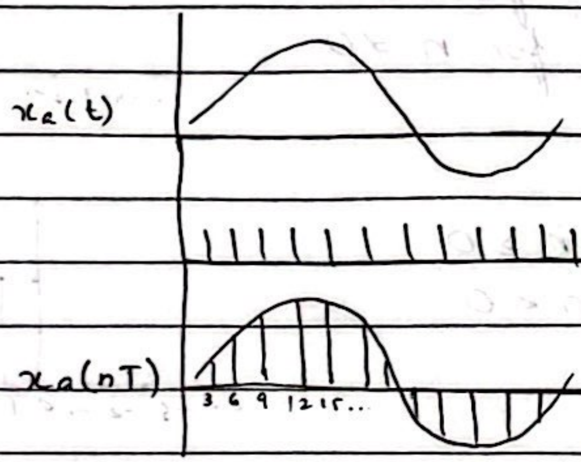
- Audio processing
- Image processing
- Speech processing
- Communication
- Sensors & Control
- Signal detection & tracking
- Video processing
- Software defined Radio

- Space
- Medical
- Commercial
- Telephone
- Military
- Industrial
- Scientific

Ch # 2 → Descrete Time Signal & Systems

2.1

(Graphical representation



here $T = 3$

Some representations are :

Functional :

$$x(n) = \begin{cases} 1 & \text{For } n = 1, 3 \\ 4 & \text{For } n = 2 \\ 0 & \text{elsewhere} \end{cases}$$

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DAY-1

Tabular :

n	... -2 -1 0 1 2 3 4 5
$x(n)$	... 0 0 0 1 4 1 0 0

Sequence :

$x(n) = \{ \ldots 0, 0, 1, 4, 1, 0, 0 \ldots \}$
↑

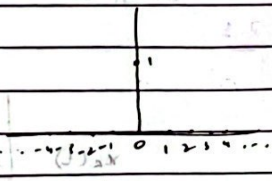
Arrow where $n=0$

DAY-2

2.1.1

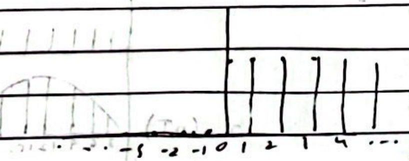
1) Unit sample sequence

$$\delta(n) = \begin{cases} 1 & \text{for } n = 0 \\ 0 & \text{for } n \neq 0 \end{cases}$$



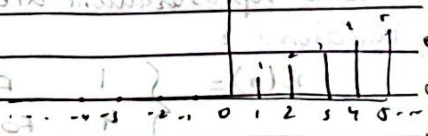
2) Unit step signal

$$u(n) = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$



3) Unit ramp signal

$$u_r(n) = \begin{cases} n & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$



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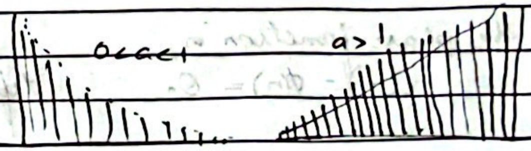
4) Exponential signal

$x(n) = a^n$ for all n



→ if a is real, then $x(n)$ is real signal

Let $n = -2 : 1 : 2$

$a = -0.5$

n	a^n
-2	4
-1	-2
0	1
1	-0.5
2	0.25

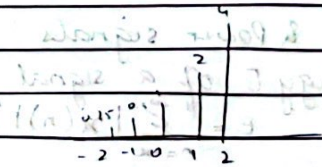
:: Example

Example

$a = 2$

$n = -2 : 2$

n	a^n
-2	0.25
-1	0.5
0	1
1	2
2	4



→ If a is a complex valued,

$a = re^{j\theta}$

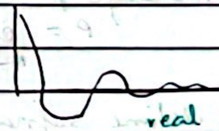
$(re^{j\theta})^n$

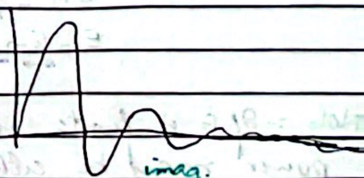
$r^n e^{j\theta n}$

$r^n(\cos\theta n + j \sin\theta n)$

$\underbrace{r^n \cos\theta n}_{\text{real}} + j\underbrace{r^n \sin\theta n}_{\text{imag}}$

given ← r, θ → vary, θ → given





The amplitude function is

$|x(n)| = A(n) = r^n$

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the phase function is

$x(n) = \phi(n) = \Theta_n$ defined over the interval $-\pi \le \phi \le \pi$

or $0 \le \Theta \le 2\pi$

Example :

$\phi(u) = \{-3\pi, -\frac{\pi}{2}, \pi, \frac{3\pi}{2}, 3\pi\}$

without normalization, wide range is (3π to 3π)
After normalization, the range is ($-\pi$ to π)

```
n =-20 : 20;
y = (pi/10) * (n) - 2 * pi * round (n/20);
```

2.1.2 Energy & Power signals

the energy E of a signal $x(n)$ is — can be finite or infinite

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

If E is finite ($0 < E < \infty$) then $x(n)$ is energy signal

the average power is

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

If we define signal energy of $x(n)$ ($-N \le n \le N$)

$$E_N = \sum_{n=-N}^{N} |x(n)|^2$$

$$E = \lim_{N\to\infty} E_N$$

Note :- If E is finite, $P = 0$. If E is infinite, the average power may be either finite or infinite. If P is finite, the signal is a power signal.

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Example : (It is a unit step sequence)

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{N} u^2(n)$$ → let $n = 2$

$u^2(0) + u^2(1) + u^2(2)$

$1 + 1 + 1 \Rightarrow 3 \Rightarrow N+1$

$$= \lim_{N\to\infty} \frac{N+1}{2N+1}$$ How

$$= \lim_{N\to\infty} \frac{1 + 1/N}{2 + 1/N} \Rightarrow \frac{1}{2}$$

it is a power signal. $E = \infty$

$Ae^{j\omega_0 n}$ Find energy & Power

DAY- 3

Periodic & aperiodic signal :

$x(n+N) = x(n)$ → periodic

Also

$x(n) = A \sin 2\pi f_0 n$

is periodic if f_0 is a rational number.

Symmetric & antisymmetric signals :

$x(-n) = x(n)$ → symmetric (even)

$x(-n) = -x(n)$ → antisymmetric (odd)

COPY

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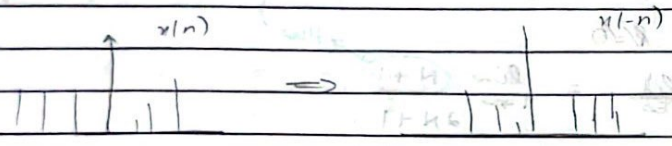
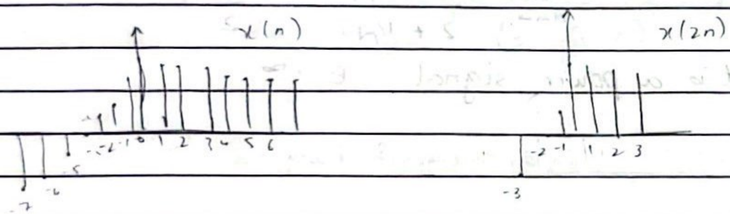
A signal $x(n)$ may be shifted in time by replacing the independant variable $n(n-k)$





DAY-3

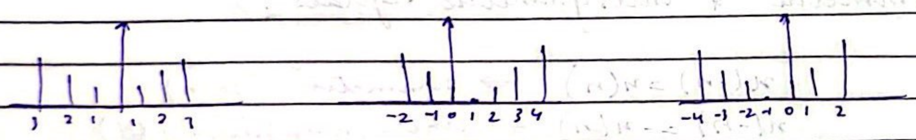
2.2. Discrete Time system:

$$y(n) = T[x(n)]$$

Example 2.2.1

$$x(n) = \begin{cases} |n| & -3 \le n \le 3 \\ 0 & \text{otherwise} \end{cases}$$

a) $y(n) = x(n)$ b) $x(n-1)$ c) $x(n+1)$



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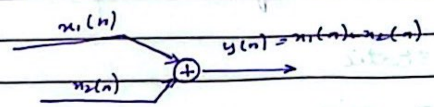
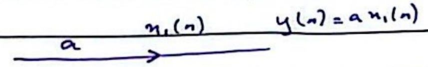
DAY-4

Block diagram Representation of DTS:

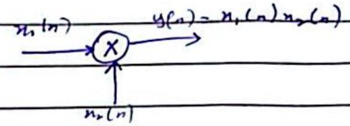
→ Adder.



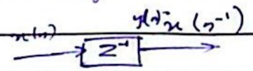
→ A constant multiplier:



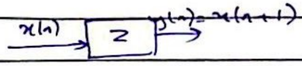
→ A signal Multiplier:



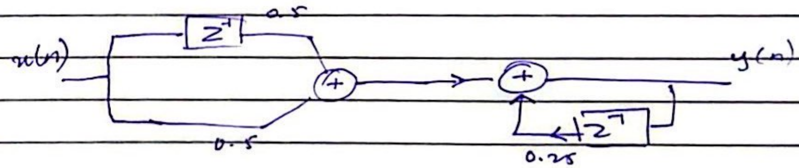
→ Unit Delay element:



→ Unit advance element.



Example 2.2.3



Classification of discrete Time System :

→ the output should not depend on input of past or future ~~value~~ samples is static system.

$$y(n) = x(n) \rightarrow \text{static}$$

$$\left.\begin{aligned} y(n) &= x(n-1) \\ y(n) &= x(n+1) \end{aligned}\right\} \text{Dynamic}$$

Example :

$$y(n) = x(-n)$$

$$y_1(n) = x_1(-n)$$

$$x_2(n) = x_1(n-k)$$

$$y_2(n) = x_1(n-k) \rightarrow (1)$$

$$y_1(n) = x_1(-n-k) \rightarrow (2)$$

$$LHS \neq RHS$$

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DAY-4

Classification of discrete Time System :

→ the output should not depend on input of past or future samples value is static system.

$y(n) = x(n)$ → static

$y(n) = x(n-1)$ } Dynamic

$y(n) = x(n+1)$

Example:

$y(n) = x(-n)$

$y_1(n) = x_1(-n)$

$x_2(n) = x_1(n-k)$

$y_2(n) = x_1(n-k)$ → ①

$y_1(n) = x_1(-n-k)$ → ②

LHS ≠ RHS

Completed Week 2.2 slides.

DAY-5

A system in which the output at any time n depends only on present and past values is called causal.

.3.2 Resolution of DTS :

$x_k(n) = \delta(n-k)$

$x(n)\delta(n-k) = x(k)\delta(n-k)$

Rest from slides

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DAY-6

$\cos[x(n)]$ → stable or not

As we know

$-1 < \cos[x(n)] < 1$

so, it is a stable system.

Correlation :

A mathematical operation which closely resembles convolution is correlation.

x → transferred signal

y → received signal

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)\,y(n-l)$$

l → Delay

End of ch# 2.

Ch # 4

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} = \begin{cases} N & k = 0, \pm N, \pm 2N \\ 0 & \text{otherwise} \end{cases}$$

DAY-7

Convolution :

Example:

Determine the output $y(n)$ of a relaxed LTI system with impulse response

$h(n) = a^n u(n)$, $|a| < 1$

unit sequence

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when the input is a unit step sequence, that is

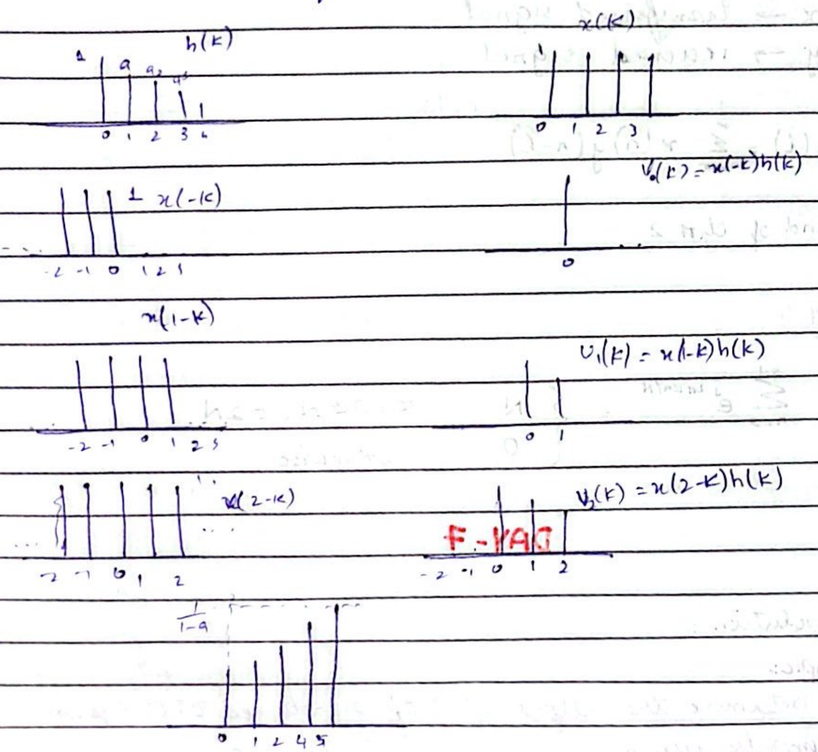
$$x(n) = u(n)$$

solution:

In this case both $h(n)$ and $x(n)$ are infinite duration sequences. We use following form of convolution formula

$$y(n) = \sum_{k=-\infty}^{\infty} x(n-k)\, h(k)$$

in which $x(k)$ is folded.



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$y(0) = 1$

$y(1) = 1 + a$

$y(2) = 1 + a + a^2$

$y(n) = 1 + a + a^2 \dots a^n$

$$= \frac{(1-a)^{n+1}}{1-a}$$

By infinity $= \lim_{n \to \infty} y(n) = \frac{1}{1-a}$

Homework:

Determine the impulse response for the cascade of two LTI system having impulse responses

$$h_1(n) = \left(\frac{1}{2}\right)^n u(n) \quad \text{and} \quad h_2(n) = \left(\frac{1}{4}\right)^n u(n)$$

DAY-8

solution:

To determine the overall response of the two systems in cascade, simply convolve $h_1(n)$ and $h_2(n)$. Hence,

$$h(n) = \sum_{k=-\infty}^{\infty} h_1(k)\, h_2(n-k)$$

where $h_2(n)$ is folded & shifted

We define the product sequence

$$V_n(k) = h_1(k)\, h_2(n-k)$$

$$= \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k}$$

which is non-zero for $k \geq 0$ & $n-k \geq 0$

On the other hand, for $n<0$, we have $v_n(k)=0$ for all k. Hence,

$$h(n) = 0, \quad n<0$$

For $n \geq k \geq 0$

$$h(n) = \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k}$$

$$= \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{2(n-k)} \qquad \therefore \left(\frac{1}{4}\right)^{n-k} = \left(\frac{1}{2^2}\right)^{n-k} = \left(\frac{1}{2}\right)^{2(n-k)}$$

$$= \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{2n-2k}$$

Since $\left(\frac{1}{2}\right)^{2n}$ is independent of $\sum_k$, factor it out

$$= \left(\frac{1}{2}\right)^{2n} \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{-2k}$$

$$= \left(\frac{1}{2}\right)^{2n} \sum_{k=0}^{n} \left(\frac{1}{2}\right)^{-k}$$

$$= \left(\frac{1}{2}\right)^{2n} \left(2^{n+1}-1\right) \qquad \therefore \sum_{k=0}^{n} \left(\frac{1}{2}\right)^{-k} = \frac{1-2^{n+1}}{1-2} = 2^{n+1}-1$$

$$= \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n \left(2^{n+1}-1\right)$$

$$= \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n \left(2^n\{2-1\}\right)$$

$$= \left(\frac{1}{2}\right)^n \left[2^{-n}2^{n}2 - 2^{-n}\right]$$

$$\boxed{h(n) = \left(\frac{1}{2}\right)^n \left[2 - \left(\frac{1}{2}\right)^n\right]}$$

DAY-9

Power Density Spectrum of Periodic Signals

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 \rightarrow \text{Average power}$$

Example:

$x(n) = [1, -2, 1, -2, 1, -2] \rightarrow$ any sequence

$$= \frac{1}{2} \sum_{n=0}^{1} |x(n)|^2$$

$$= \frac{1}{2} \left[|x(0)|^2 + |x(1)|^2\right]$$

$$= \frac{1}{2} \left[|1|^2 + |-2|^2\right]$$

$$= \frac{1}{2} \left[1 + 4\right]$$

$$\boxed{P_x = \frac{5}{2}}$$

$$E_N = \sum_{n=0}^{N-1} |x(n)|^2 = N \sum_{k=0}^{N-1} |c_k|^2 \rightarrow \text{Energy}$$

Example 4.2.2 :

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} = \frac{1}{N} \sum_{n=0}^{L-1} A e^{-j2\pi kn/N}$$

$$c_k = \frac{A}{N} \sum_{n=0}^{L-1} \left(e^{-j2\pi k/N}\right)^n = \begin{cases} AL/N & k=0 \\ \frac{A}{N} \frac{1-e^{-j2\pi kL/N}}{1-e^{-j2\pi k/N}}, & k \neq 0 \end{cases}$$

$$\frac{1-e^{-j2\pi kL/N}}{1-e^{-j2\pi k/N}} = \frac{e^{-j\pi kL/N}}{e^{-j\pi k/N}} \cdot \frac{e^{j\pi kL/N} - e^{-j\pi kL/N}}{e^{j\pi k/N} - e^{-j\pi k/N}}$$

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P-VAD

$$C_k = \begin{cases} \frac{AL}{N} & k=0, \pm N, \pm 2N \\ \frac{A}{N} e^{-} & \end{cases}$$

Example 4.2.3

$x(n) = a^n u(n) \qquad -1<a<1$

Σ

DAV-10

Find DTFT of following signal:

$$x(n) = \begin{cases} 1 & 0 \le n \le 4 \\ 0 & \text{otherwise} \end{cases}$$

sol

$$X(\omega) = \sum_{n=0}^{4} x(n) e^{-j\omega n}$$

$$= x(0)e^{0} + x(1)e^{-j\omega} + \dots \; x(4)e^{-j4\omega}$$

$$X(\omega) = 1 + e^{-j\omega} + e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega}$$

We know $X(\omega) = |x(\omega)| e^{j\theta}$

Lets find $|x(\omega)|$ & $e^{j\omega\theta}$

$$\sum_{n=0}^{4} x(n) e^{-j\omega n} = \sum_{n=0}^{4} e^{-j\omega n} = \sum_{n=0}^{N-1} (e^{-j\omega})^n \qquad \text{So } n=5$$

$$= \frac{1-e^{-j\omega 5}}{1-e^{-j\omega}} = \frac{e^{-j\omega 5/2}}{e^{-j\omega/2}} \cdot \frac{e^{j\omega 5/2} - e^{-j\omega 5/2}}{e^{j\omega/2} - e^{-j\omega/2}}$$

$$= e^{-2j\omega} \cdot \frac{\sin 5\omega/2}{\sin \omega/2} = e^{j\theta} \cdot |x(\omega)|$$

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As

$$e^{-2j\omega} = e^{j\theta}$$

Hence

$$\angle X(\omega) = -2\omega$$

$$|X(\omega)| = \frac{\sin 5\omega/2}{\sin \omega/2}$$

Find DTFT of following signal:

$$x(n) = \{ a^n u(n) \qquad |a|<1$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n u(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (a e^{-j\omega})^n$$

$\therefore \sum_{k=a}^{\infty} x^k = \{ \frac{x^a}{1-x}$

$$= \frac{(a e^{-j\omega})^0}{1 - a e^{-j\omega}} = \frac{1}{1 - a e^{-j\omega}}$$

$$|X(\omega)| = \frac{1}{\sqrt{(1-a\cos\omega)^2 + (a\sin\omega)^2}} = \frac{1}{\sqrt{1^2 + a^2\cos^2\omega - 2a\cos\omega + a^2\sin^2\omega}}$$

$$= \frac{1}{1 - 2a\cos\omega + a^2(\cos^2\omega + \sin^2\omega)}$$

$$= \frac{1}{\sqrt{1 - 2a\cos\omega + a^2}}$$

Lets find minima and maxima (for plotting) by derivative

Let $g(\omega) = 1 - 2a\cos\omega + a^2$

$$\frac{dg}{d\omega} = 0 - 2a(-\sin\omega) + a = 0$$

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$g(\omega) = 2a\sin\omega = 0 \Rightarrow \sin\omega = 0$

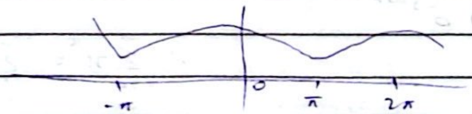
sin is zero at 0, π + 2π

$$|X(0)| = \frac{1}{\sqrt{1-2a+a^2}} = \frac{1}{\sqrt{(1-a)^2}}$$

$$|X(0)| = \frac{1}{1-a}$$

$$|X(2\pi)| = \frac{1}{1-a}$$

$$|X(\pi)| = \frac{1}{\sqrt{1+2a+a^2}} = \frac{1}{\sqrt{(1+a)^2}} = \frac{1}{1+a}$$

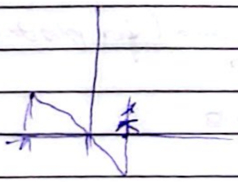


$$\angle X(\omega) = \angle 1 - \angle(1 - a\cos\omega + ja\sin\omega)$$

$$= \tan^{-1}\left(\frac{0}{1}\right) - \tan^{-1}\left(\frac{a\sin\omega}{1-a\cos\omega}\right)$$

$$= 0 - \tan^{-1}\left(\frac{a\sin\omega}{1-a\cos\omega}\right)$$

$$= \cancel{0} - \tan^{-1}\left(\frac{a\sin\omega}{1-a\cos\omega}\right)$$



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Example

$$x(n) = \begin{cases} A & 0 \le \\ 0 & \end{cases}$$

$$= A\,\frac{1-e^{-j\omega L}}{1-e^{-j\omega}}$$

$X(\omega) =$

Homework:

DTFT

→ Abstract } google scholar

→ Conclusion

DAY-11

Q.1

Find DTFT of following signal.

$$x(n) = \begin{cases} 1 & 0 \le n \le 4 \\ 0 & \text{otherwise} \end{cases}$$

Sol

$$x(\omega) = \sum_{n=0}^{4} x(n)\,e^{-j\omega n}$$

$$= x(0)e^{0} + x(1)e^{-j\omega} + \dots x(4)e^{-j4\omega}$$

$$X(\omega) = 1 + e^{-j\omega} + e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega}$$

we know

$$X(\omega) = |X(\omega)|\,e^{j\theta}$$

Let's find $|x(\omega)|$ & $e^{j\omega\theta}$

$$= \sum_{n=0}^{4} e^{-j\omega n} = \sum_{n=0}^{N-1} (e^{-j\omega})^n \quad \text{so } N = 5$$

$$= \frac{1-e^{-j\omega 5}}{1-e^{-j\omega}} = \frac{e^{-j\omega 5/2}}{e^{-j\omega/2}} \cdot \frac{e^{j\omega 5/2} - e^{-j\omega 5/2}}{e^{j\omega/2} - e^{-j\omega/2}}$$

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$$= e^{-2jw} \cdot \frac{\sin 5w/2}{\sin w/2} = e^{j\theta} \cdot |x(w)|$$

Q.2

Find DTFT of following signal

$$x(n) = \{ a^n u(n) \quad , |a| < 1$$

$$X(w) = \sum_{n=-\infty}^{\infty} x(n) e^{-jwn} = \sum_{n=0}^{\infty} a^n e^{-jwn}$$

$$= \sum_{n=0}^{\infty} (a e^{-jw})^n$$

$$= \frac{(a e^{-jw})^0}{1 - a e^{-jw}} = \frac{1}{1 - a e^{-jw}}$$

$$= \frac{1}{1 - a(\cos w - j \sin w)}$$

$$= \frac{1}{1 - a \cos w + j\, a \sin w}$$

$$|X(w)| = \frac{1}{\sqrt{(1 - a\cos w)^2 + (a \sin w)^2}} = \frac{1}{\sqrt{1^2 + a^2 \cos^2 w - 2a\cos w + a^2 \sin^2 w}}$$

$$= \frac{1}{\sqrt{1 - 2a\cos w + a^2(\cos^2 w + \sin^2 w)}} = \frac{1}{\sqrt{1 - 2a\cos w + a^2}}$$

Lets find minima & maxima (for plotting) by derivation

Let $g(w) = 1 - 2a \cos w + a^2$

$$\frac{dg}{dw} = 0 - 2a(-\sin w) + 0 = 0$$

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$$g(w) = 2a \sin w = 0$$

$$\sin w = 0$$

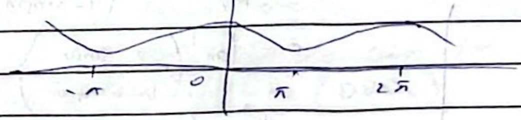
sin is zero at $0, \pi + 2\pi$

$$|X(0)| = \frac{1}{\sqrt{1 - 2a + a^2}} = \frac{1}{\sqrt{(1-a)^2}}$$

$$|X(0)| = \frac{1}{1-a}$$

$$|X(2\pi)| = \frac{1}{1-a}$$

$$|X(\pi)| = \frac{1}{\sqrt{1 - 2a + a^2}} = \frac{1}{\sqrt{(1+a)^2}} = \frac{1}{1+a}$$



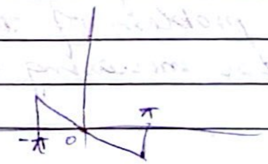
$$\angle x(w) = \angle num - \angle den$$

$$= \angle 1 - \angle (1 - a\cos w + j\, a \sin w)$$

$$= \tan^{-1}\left(\frac{0}{1}\right) - \tan^{-1}\left(\frac{a \sin w}{1 - a\cos w}\right)$$

$$= 0 - \tan^{-1}\left(\frac{a \sin w}{1 - a \cos w}\right)$$

$$= -\tan^{-1}\left(\frac{a \sin w}{1 - a \cos w}\right)$$



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Example 4.4.4 :

$y(n) = ay(n-1) + bx(n) \quad 0 < a < 1$

a) Determine magnitude and phase

$h(n) = ba^n u(n)$

$H(\omega) = \sum_{n=-\infty}^{\infty} h(n) e^{-j\omega n}$

$H(\omega) = \dfrac{b}{1-ae^{-j\omega}} = \dfrac{b}{(1-a\cos\omega) + \sin j\omega}$

$\angle H(\omega) = \angle num - \angle den$

$= \angle b - \angle(1-ae^{-j\omega}) \Rightarrow \tan^{-1}\left(\dfrac{0}{b}\right) - \tan^{-1}\left(\dfrac{a\sin\omega}{1-a\cos\omega}\right)$

$|H(\omega)| = \dfrac{b}{\sqrt{(1-a\cos\omega)^2 + (a\sin\omega)^2}}$

$= \dfrac{b}{\sqrt{1-2a\cos\omega + a^2(\cos^2\omega + \sin^2\omega)}}$

$|H(\omega)| = \dfrac{b}{\sqrt{1+a^2-2a\cos\omega}}$

$0 - \tan^{-1}\left(\dfrac{a\sin\omega}{1-a\cos\omega}\right)$

$= -\tan^{-1}\left(\dfrac{a\sin\omega}{1-a\cos\omega}\right)$

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Multi Rate Digital Signal Processing

In many practical applications of digital signal processing one is faced with the problem of changing the sampling rate of a signal, either increasing it or decreasing it.

For Example:

i) In Telecommunication system, that transmit and receive different types of signal (e.g speak, video etc) There is a requirement to process the various signals at different rates according to the corresponding bandwidth of the signals at the process of converting a signal from a given rate to a different rate is called sampling rate conversion. In turn, systems that employ multiple sampling rates in the processing of digital signals are called multi rate digital signal processing system.

ii) Audio Processing:

Audio signals can be converted from 44.1KHz (for CDs) to 48KHz (DVDs)

iii) Image Processing:

Images/videos can be upscaled to higher resolution

iv) Seismic Data Processing.

The size of seismic data can be reduced for storage and transmission.

v) Reduced Bandwidth:

By reducing the sampling rate of a signal (Decimation) the amount of data that needs to be transmitted / stored is decreased at this directly reduces the required bandwidth.

$$x(n) \rightarrow \boxed{\text{system}} \rightarrow y(n)$$

$F_x = \frac{1}{T_x}$ $\quad F_y = \frac{1}{T_y}$

$F_y \neq F_x$

Sampling rate conversion of a digital signal can be accomplished in one of two general methods

→ One method is to pass the digital signal through a D/A converter, filter it if necessary and then to resample the resulting analog signal at the desired rate. The second method is to perform the sampling rate conversion entirely in the digital domain.

1) First method :

$$x(n) \rightarrow \boxed{\text{D/A convertor}} \rightarrow \boxed{\text{sampler}} \rightarrow y(n)$$

$F_x = \frac{1}{T_x}$ $\quad T_x \neq T_y$ $\quad F_y = \frac{1}{T_y}$

Pros and cons :

* the benefit of this method is that F_x and F_y can have any arbitrary value.
* the problem in this method is extensive approximation, which distorts the signal.

ii) Digital Domain method :

* In this method, sampling rate conversion is performed entirely in the digital domain.
* In this method, F_x and F_y will have some relationship. However, we need to perform D/A conversion.

$F_y = I F_x$

Up sampler :

$$x(n) \rightarrow \boxed{\text{Discrete system}} \rightarrow y(n)$$

$F(x)$ $\quad F_y$

$F_y > F_x$

Let

$x(n) = \{1 \; 2 \; 3 \; 4\}$

$n = 0 \; 1 \; 2 \; 3$

$$F_x \rightarrow \boxed{\uparrow I} \rightarrow F_y$$

$F_y = I F_x$

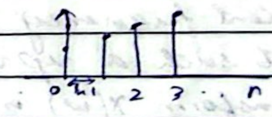
$(I-1)$ zeros will be added between points.

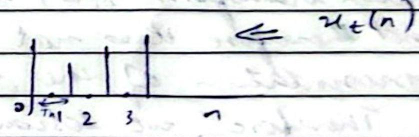
If sampling rate is increased, it means T_x is reduced & F_x is increased.

Lets assume we upsample the signal by a factor of 2.

$$x(n) \rightarrow \boxed{\uparrow 2} \rightarrow y(n)$$

$F_y = 2F_x$





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Down Sampler :

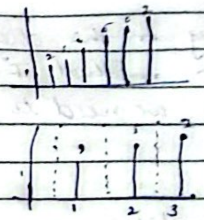
$$F_x \rightarrow \boxed{\downarrow 2} \rightarrow F_y$$

$F_x > F_y$

Let

$x(n) = \{ 1\ 2\ 3\ 4\ 5\ 6\ 7 \}$

$n:\ 0\ 1\ 2\ 3\ 4\ 5\ 6$



Decimation by Factor D:

$$F_x \rightarrow \boxed{\downarrow D} \rightarrow F_y$$

$$F_y = \frac{F_x}{D}$$

(D-1) samples will be dropped.

Ch # 7 → Discrete Fourier Transform

Frequency analysis of a discrete time signals is usually performed on a digital signal processor. To perform frequency analysis on a discrete-time signal $\{x(n)\}$, we convert the time domain sequence to an equivalent frequency domain representation. We know that such a representation is given by the fourier transform $X(\omega)$ is continuous function of frequency and therefore it is not a computationally convenient representation of the sequence $\{x(n)\}$.

Therefore, we know consider the representation of a sequence $x(n)$ by samples of its spectrum $X(\omega)$. Such a representation is called discrete Fourier Transform (DFT).

down sampler

- → reducing samples can speed up the processing time
- → reduces the space of samples.

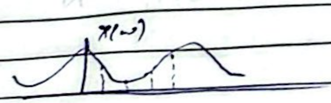
→ If $X(\omega)$ is continuous, it is DTFT and if $X(\omega)$ is discrete, it is DFT.

7.1 Frequency-domain sampling: The DFT

7.1.1 Frequency domain sampling of DT signals.

Recall that

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n} \rightarrow 1.1$$



Suppose that we sample $X(\omega)$ periodically in frequency at spacing of $\delta\omega$ radians between successive samples, where $X(\omega)$ is periodic with period 2π.

=> For convenience, we take N equivalent samples in the interval $0 \le \omega < 2\pi$ with spacing $\delta\omega = \frac{2\pi}{N}$, as shown in figure

=> If we evaluate (7.1.1) at $\omega = \frac{2\pi k}{N}$, we obtain

$$X\left(\frac{2\pi k}{N}\right) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\frac{2\pi k n}{N}} \qquad k = 0, 1, \dots, N-1$$

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Discrete Fourier Transform:

DFT:

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi k n}{N}} \rightarrow ① \qquad k = 0, 1, \dots, N-1$$

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IDFT:

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)\, e^{\frac{j2\pi kn}{N}} \rightarrow ② \quad n = 0,1,2, \dots N-1$$

The DFT as a linear Transformation:

The formulas in eq ① & eq ② may be expressed as

$$X(k) = \sum_{n=0}^{N-1} x(n)\, W_N^{kn} \qquad k = 0,1,\dots N-1$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)\, W_N^{-kn} \qquad n = 0,1,\dots N-1$$

where

$$W_N = e^{-j2\pi/N}$$

4-Point DFT:

$N = 4$, $k = n = 0:3$

For

$k=0 \Rightarrow x(0)W_4^0 + x(1)W_4^0 + x(2)W_4^0 + x(3)W_4^0$

$k=1 \Rightarrow x(0)W_4^0 + x(1)W_4^1 + x(2)W_4^2 + x(3)W_4^3$

$k=2 \Rightarrow x(0)W_4^0 + x(1)W_4^2 + x(2)W_4^4 + x(3)W_4^6$

$k=3 \Rightarrow x(0)W_4^0 + x(1)W_4^3 + x(2)W_4^6 + x(3)W_4^9$

$\Downarrow$

$W_N x_N = X(k)$ ← 4 point DFT

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$$W_N = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}$$

$W_4^0 = ?$
$W_4^1 = ?$
$W_4^2 = ?$
$W_4^3 = ?$

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$W_4^0 = W_4^{kn} = e^{-j\frac{2\pi(0)}{4}} = 1$

$W_4^1 = e^{-j2\pi/4} = e^{-j\pi/2} = \cos\left(\frac{\pi}{2}\right) - j\sin\left(\frac{\pi}{2}\right) = 0 - j(1) = -j$

$W_4^2 = e^{-j\frac{2\pi 2}{4}} = e^{-j\pi} = \cos(\pi) - j\sin(\pi) = -1 - 0 = -1$

$W_4^3 = e^{-j\frac{2\pi 3}{4}} = e^{-j3\pi/2} = \cos\left(\frac{3\pi}{2}\right) - j\sin\left(\frac{3\pi}{2}\right) = 0 - j(-1) = j$

Example:

Compute the DFT of the 4-point sequence

$x(n) = [0 \ 1 \ 2 \ 3]$

$W_{4\times4}\, x_{4\times1}$

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ -2+2j \\ -2 \\ -2j \end{bmatrix} = X(k)$$

By periodicity property:

$W_4^0 = W_4^4$

$W_4^1 = W_4^5 = W_4^9$ $\quad 4+1=5,\ 5+4=9$

$W_4^2 = W_4^6$

$W_N^R = W_N^{R+N}$

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5.1 Frequency Domain Sampling and Reconstruction of Discrete - Time signals

Forier Transform $\leftarrow$ $X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$ $\rightarrow$ no. of samples

From slides

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Circular Symmetries of a Sequence :

As we have seen, the N-point DFT of a sequence, $x(n)$ of length $N \geq L$ is equivalent to the N-point DFT of a periodic sequence $x_p(n)$, of period N, which is obtained by periodically extending $x(n)$, that is

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n - lN)$$

Now suppose we shift the periodic sequence $x_p(n)$ by 'k' units to the right. Thus we obtain another periodic sequence.

$$x'_p(n) = x_p(n-k) = \sum_{l=-\infty}^{\infty} x(n-k-lN)$$

The finite - duration sequence

$$x'(n) = \begin{cases} x'_p(n) & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

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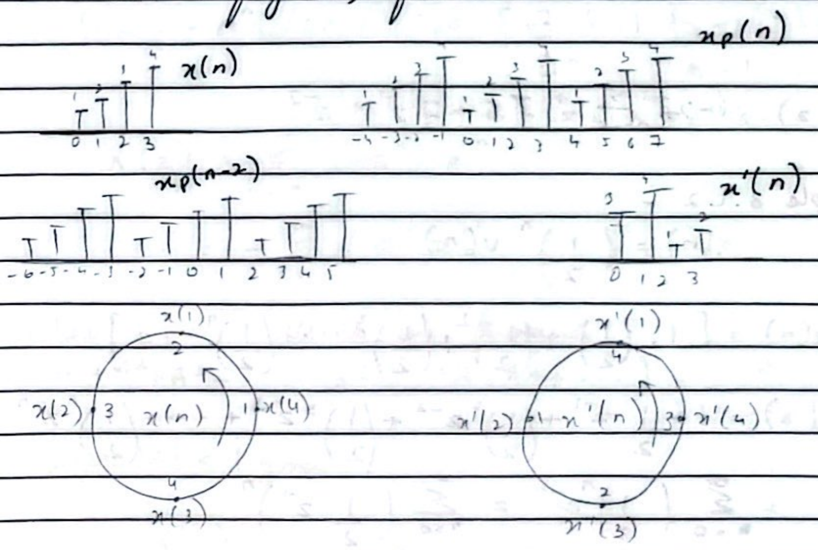
is related to the $x(n)$ by a circular shift As shown in fig 1, for N = 4



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Chapter 3 -

Transform techniques are an important tool in the analysis of signals and linear time-invariant systems.

3.1 The Z-transform :

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

$$x(n) \overset{z}{\longleftrightarrow} X(z)$$

1) $x_1(n) = \{1, 2, 5, 7, 0, 1\}$

$$= (1)z^{0} + 2z^{-1} + 5z^{-2} + 7z^{-3} + 1z^{-5}$$

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$$X_1(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$

b) $x_2(n) = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

$$X_2(z) = 1z^{+2} + 2z^{+1} + 5 + 7z^{-1} + z^{-3}$$

Example 3.1.2

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$x(n) = \left[1, \left(\frac{1}{2}\right), \left(\frac{1}{2}\right)^2, \left(\frac{1}{2}\right)^3 \dots \left(\frac{1}{2}\right)^n \dots\right]$$

$$X(z) = 1 + \frac{1}{2}z^{-1} + \left(\frac{1}{2}\right)^2 z^{-2} + \left(\frac{1}{2}\right)^3 z^{-3} + \dots \left(\frac{1}{2}\right)^n z^{-n} \dots$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} = \sum_{n=0} \left(\frac{1}{2} z^{-1}\right)^n$$

$$1 + A + A^2 + A^3 + \dots \quad \frac{\quad}{1 - A} \quad \text{if } |A| < 1$$

$$\left|\frac{1}{2} z^{-1}\right| < 1$$

$$X(z) = \frac{1}{1 - \frac{1}{2} z^{-1}} \qquad ROC: |z| > \frac{1}{2}$$

Example 3.1.3

$$x(n) = a^n u(n) = \begin{cases} a^n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

$$x(n) = a^n u(n) \overset{z}{\longleftrightarrow} X(n) = \frac{1}{1 - az^{-1}} \qquad ROC: |z| > a$$

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Example 3.1.4

$$x(n) = -a^n u(-n-1) = \begin{cases} 0 & n \geq 0 \\ -a^n & n \leq -1 \end{cases}$$

$$X(z) = \sum_{n=-\infty}^{-1} (-a^n) z^{-n}$$

$$= -\sum_{l=1}^{\infty} (a^{-1} z)^l$$

where $l = -n$ using the formula

$$A + A^2 + A^3 + \dots = A(1 + A + A^2 + \dots) = \frac{A}{1 - A}$$

where $|A| < 1$ gives

$$X(z) = -\frac{a^{-1} z}{1 - a^{-1} z} = \frac{1}{1 - az^{-1}}$$